Total No. of Questions : 22

Total No. of Pages : 02

Roll No. ______

B.Tech IV sem (Main/Back) Exam April-May 2026
Artificial Intelligence & Data Science
4AID4-06 Theory of Computation
Paper Code : 4E1306
CS, IT, AID, CAI, CCS, CDS, CIT,CSR

Time: 3 Hours

Maximum Marks: 70

Attempt all ten questions from Part A, five questions out of seven questions from Part B and three questions out of five questions from Part C.

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. Units of quantities used/ calculated must be stated clearly.

Use of following supporting material is permitted during examination. (Mentioned in form No.205)

1 ______ 2. ______

Part A (Answer should be given up to 25 words only)

All questions are compulsory

Q.1 Define a FA which checks whether the given binary number in even.

Q.2 Give application of pumping lemma.

Q.3 State parse tree with example.

Q.4 Show that the following grammar is ambiguous or not : $S \rightarrow aSbS / bSaS / \varepsilon$.

Q.5 Write three properties of CFL.

Q.6 What is the difference between finite automata and pushdown automata ?

Q.7 Define the recursive and recursively enumerable languages.

Q.8 Prove the transpose closure property of context sensitive language.

Q.9 Define NP complete and NP hard problem.

Q.10 Define Cook's theorem and its application.

10 x 2 = 20

Part B(Analytical/Problem solving questions)

Attempt any Five questions

Q.1 Convert the following Moore machine into Melay machine -

State	Input a	Input b	output
$\rightarrow q_0$	q_1	q_3	1
q_1	q_3	q_1	0
q_2	q_0	q_3	0
q_3	q_3	q_2	1

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Contd

Q.2 Construct the finite automation equivalent to the regular expression **(a+b)* a b * a (a+ba)***

Q.3 Explain context free grammar. Also differentiate it from context sensitive language.

Q.4. Convert the following context free Grammar (CFG) into a equivalent CFG in Chomsky normal form :

$S \to ASA|aB$

$A \to B|S$

$B \to b|\varepsilon$

Q.5 Design PDA for accepting $\{a^n b^n c^m d^m \mid n, m \geq 1\}$.

Q.6 Design a Turning machine M to recognize the language $\{a^n b^n \mid n \geq 1\}$.

Q.7 Explain travelling salesman problem with example.

5 x 4 = 20

Part C (Descriptive/Analytical/Problem Solving/Design question)

Attempt any three questions

Q.1 *Differentiate* between deterministic and non deterministic finite automation. Find the equivalent DFA for the NFA given by M=[{A, B, C}, (a,b), δ, A, {c}] where δ is given by :

	A	b
→A	A , B	C
B	A	B
C	-	A, B

Q.2 Given grammer G

S → a B | bA

A → a |aS|bAA

B → b | bS | aBB

Derive the string **"baaabbabba"** find left most derivation, right most derivation and parse tree.

Q.3 How can a pushdown automata be constructed for a given language ? Explain with example.

Q.4 Explain Chomsky classification of language with the help of suitable example.

Q.5 Write algorithm for approximation for vertex cover problem with suitable example.

3 x 10= 30